Lecture 2

Qubits and interference

Objectives. *Lecture 1 computed with bits, sure or random, by products of matrices in sequence and Kronecker products in parallel. This lecture keeps those rules and makes the entries complex amplitudes; a probability becomes the squared modulus of a sum of them. It asks:* what changes when probabilities become amplitudes? *The path sum (Theorem 2.3) answers: an amplitude is a sum over paths; the squared moduli of path amplitudes give a probabilistic circuit of Lecture 1, and the cross terms are interference. Its first visible consequence is Deutsch's problem, which one quantum query solves and one classical query cannot.*

By the end of this lecture you will be able to:

- Translate the spin $\frac{1}{2}$ preparations and measurements of QM-I into circuit notation, and compute the probabilities of a reading.
- Read a circuit of gates as one unitary matrix, and its mirror image, with adjoint boxes, as the inverse.
- Write an output amplitude as a sum over paths, and split its probability into the part a probabilistic circuit gives and the interference.
- Solve Deutsch's problem with one query by phase kickback, prove that one classical query cannot, and say why the control qubit changes.

2.1 From QM-I to circuit notation

Quantum Mechanics I (QM-I), Lecture 1, already calls a spin $\frac{1}{2}$ a qubit. We keep its states and measurement rules. The table records the changes of notation and of the role of a matrix:

QM-I	This course
$\alpha\,\lvert 0\rangle + \beta\,\lvert 1\rangle$	$\alpha\,\lvert 0\rangle + \beta\,\lvert 1\rangle$
$\lvert 0\rangle$, z reading $+1$	$\lvert 0\rangle$, with $Z\lvert 0\rangle = \lvert 0\rangle$
$\lvert 1\rangle$, z reading -1	$\lvert 1\rangle$, with $Z\lvert 1\rangle = -\lvert 1\rangle$
observables $\sigma_x, \sigma_y, \sigma_z$	gates X, Y, Z
$\lvert \pm x\rangle$	$\lvert \pm\rangle$
Hamiltonian evolution U_t	a gate: a unitary U
projectors E_a	projectors Π_a

QM-I uses σ_x as an observable. The same matrix is the gate X here. Applying it exchanges the basis states: $X\lvert 0\rangle = \lvert 1\rangle$ and $X\lvert 1\rangle = \lvert 0\rangle$, the NOT of Lecture 1.

In QM-I and *Quantum Mechanics II* (QM-II), a Hamiltonian fixes evolution in time. A circuit gate is specified by its matrix, so $\hbar$ does not appear in a circuit.

As in QM-I, a matrix with $A^\dagger = A$ is called *self-adjoint*. Nielsen and Chuang use "hermitian" for the same finite-dimensional condition. Here $A^\dagger$ is the conjugate transpose. The inner product is antilinear in its first slot.

Besides $X = \sigma_x$ and $Z = \sigma_z$, the course uses the Hadamard gate

$$\begin{aligned} H &= \frac{X+Z}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \end{aligned} \tag{2.1}$$

The gates X, Y, Z and H are self-adjoint and unitary. Each is also an observable, squares to $\mathbb{1}$, and is its own inverse. The Hadamard gate sends $|0\rangle$ and $|1\rangle$ to

$$|\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}, \tag{2.2}$$

the eigenvectors of X with the eigenvalues ± 1, called $|\pm x\rangle$ in QM-I, Lecture 1.

Definition 2.1 (Qubits, gates, reading)**.** A register of n *qubits* is in a *state*: a unit vector

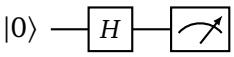
$$\psi = \sum_x \psi_x |x\rangle \in \mathbb{C}^{2^n},$$

on the columns $|x\rangle$, $x \in \{0,1\}^n$, of Lecture 1; the complex numbers ψ_x are its *amplitudes*. A *gate* on k qubits is a unitary $2^k \times 2^k$ matrix, drawn as a box and placed on its wires by the rules of Lecture 1: Kronecker products down a column, with $\mathbb{1}$ on an idle wire, and the columns multiplied right to left. *Reading* the register gives the string x with probability

$$|\langle x|\psi\rangle|^2 = |\psi_x|^2,$$

and is drawn as a meter at the end of the wires.

Reading uses the Born rule of QM-I, Lectures 1 and 4, for an observable with the eigenvectors $|x\rangle$ and 2^n different eigenvalues, so no outcome is degenerate. Since $\|U\psi\|^2 = \langle \psi | U^\dagger U \psi \rangle = 1$, the probabilities of a reading after any circuit add up to 1. In this lecture a reading ends the circuit; what it leaves behind, and the reading of part of a register, are Lecture 3's. For example,



gives 0 and 1 with probability $\frac{1}{2}$ each, since $H|0\rangle = |+\rangle$: one Hadamard, read, is a fair coin.

2.2 Soundness

Theorem 2.2 (Soundness)**.** *For complex matrices, the interchange law* $(A \otimes B)(C \otimes D) = AC \otimes BD$ *holds whenever* AC *and* BD *are defined, and*

$$\begin{aligned} (AB)^\dagger &= B^\dagger A^\dagger, \\ (A \otimes B)^\dagger &= A^\dagger \otimes B^\dagger. \end{aligned} \tag{2.3}$$

Hence, for a circuit drawn without crossings, each box on neighbouring wires:

(i) it has one matrix, however its diagram is cut into columns;

(ii) a part of it that fills a rectangle, some neighbouring wires between two cuts with

no box leaving it, may be replaced by any diagram with the same matrix, and the matrix of the whole does not change;

(iii) its mirror image, with each box U replaced by $U^\dagger$, has the adjoint matrix. So a circuit of gates is unitary, and its mirror image with adjoint boxes is its inverse.

Proof. The interchange law is Theorem 1.3, whose proof only multiplies and adds entries, and so holds over $\mathbb{C}$ as written. For the adjoints, entry by entry, with the labels of Definition 1.2,

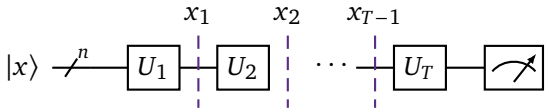
$$\begin{aligned}\big((AB)^\dagger\big)_{ik} &= \overline{\textstyle\sum_j A_{kj}B_{ji}} \\ &= \textstyle\sum_j (B^\dagger)_{ij}(A^\dagger)_{jk},\end{aligned}$$

and the entry of $(A\otimes B)^\dagger$ in row (i,i') and column (j,j') is $\overline{A_{ji}B_{j'i'}} = (A^\dagger)_{ij}(B^\dagger)_{i'j'}$. Then (i) is the argument of § 1.3. For (ii), cut at the sides of the rectangle. Before the replacement and after it, each column between the cuts is $R_t \otimes M_t \otimes R'_t$, with M_t the part's boxes and R_t, R'_t the boxes above and below it, and by the interchange law these columns multiply to $R\otimes M\otimes R'$, where M is the matrix of the part. For (iii), the adjoint of a product $K_T\cdots K_1$ of columns is $K_1^\dagger\cdots K_T^\dagger$, and each $K_t^\dagger$ is the Kronecker product of the adjoint boxes: the columns of the mirror image, read left to right. Last, $(U\otimes V)^\dagger(U\otimes V) = U^\dagger U\otimes V^\dagger V = \mathbb{1}$, so each column of gates is unitary, and so is their product. ∎

These are cases of a general theorem. Complex matrices with $\otimes$ and $\dagger$ form a dagger symmetric monoidal category, and in any such category two circuit diagrams with the same boxes wired the same way, crossings allowed, denote the same morphism, and the dagger of a diagram is its mirror image (Selinger, Thm 3.12, after Joyal and Street, and §7.2).

2.3 The path sum

To see where amplitudes and probabilities part, expand the matrix of a circuit into its entries. Cut a circuit on n qubits into its columns $U_1,\dots,U_T$, in the order in which they act:



A *path* from x to y puts a string on each cut: it is $p = (x_0, x_1, \dots, x_T)$ with $x_0 = x$ and $x_T = y$. Its *amplitude* is the product of the entries met on the way,

$$a(p) = \prod_{t=1}^{T} (U_t)_{x_t x_{t-1}}.$$

For a matrix M, write $|M|^{\circ 2}$ for the matrix of the squared moduli of its entries.

Theorem 2.3 (The path sum). *Let a circuit on n qubits have the columns $U_1,\dots,U_T$, with $T \geq 1$, and let $x, y \in \{0,1\}^n$.*

(i) The amplitude of y is the sum over the $2^{n(T-1)}$ paths from x to y:

$$\langle y|\, U_T\cdots U_1\, |x\rangle = \sum_p a(p). \tag{2.4}$$

(ii) Each $|U_t|^{\circ 2}$ is a stochastic matrix, and the probabilistic circuit with the gates $|U_1|^{\circ 2},\dots,|U_T|^{\circ 2}$, run on x, outputs y with probability $\sum_p |a(p)|^2$.

(iii) The circuit, run on $|x\rangle$ and read, outputs y with probability

$$\begin{aligned}&\left|\sum_p a(p)\right|^2\\ &\quad=\sum_p |a(p)|^2\\ &\qquad+\sum_{p\neq q}\overline{a(p)}\,a(q).\end{aligned}\tag{2.5}$$

(iv) For all matrices, $|A\otimes B|^{\circ 2}=|A|^{\circ 2}\otimes|B|^{\circ 2}$; but $|AB|^{\circ 2}$ and $|A|^{\circ 2}\,|B|^{\circ 2}$ differ in general.

Proof. (i) The entry (y,x) of a product of T matrices is the sum, over the $T-1$ indices in between, of the products of entries.

(ii) The column x of a unitary U is $U|x\rangle$, a unit vector:

$$\sum_y \left|U_{yx}\right|^2=\langle x|\,U^\dagger U\,|x\rangle=1.$$

So $|U|^{\circ 2}$ has non-negative entries and columns summing to 1: it is stochastic (Definition 1.5). The probabilistic circuit outputs y with the probability $(|U_T|^{\circ 2}\cdots|U_1|^{\circ 2})_{yx}$, and expanding this product as in (i) gives $\sum_p\prod_t\left|(U_t)_{x_t x_{t-1}}\right|^2=\sum_p|a(p)|^2$.

(iii) By Definition 2.1 and (i), the probability is $\left|\sum_p a(p)\right|^2=\sum_{p,q}\overline{a(p)}\,a(q)$, and the terms with $p=q$ are the first sum.

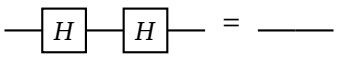
(iv) Entry by entry, $\left|A_{ij}B_{i'j'}\right|^2=\left|A_{ij}\right|^2\left|B_{i'j'}\right|^2$. For products, Example 2.4 gives $A=B=H$. ∎

By (iv), and since $|\mathbb{1}|^{\circ 2}=\mathbb{1}$, each $|U_t|^{\circ 2}$ is computed box by box. So every circuit has a *probabilistic twin*: the same diagram, each box G replaced by the stochastic matrix $|G|^{\circ 2}$, with the same paths. The twin's probability of y is the sum of the squared moduli of the path amplitudes. The quantum probability adds the cross terms of (2.5), the *interference*. Their sum is real, since the terms for (p,q) and (q,p) are conjugate, and it can be negative. A register of the twin already takes 2^n numbers to describe, as Lecture 1's foil showed, so the count is not the difference; the cross terms are.

A run of the circuit returns one string y, drawn with the probability (2.5). The sum over $2^{n(T-1)}$ paths is how that probability is computed, not something the circuit evaluates and hands over: no run reads all the paths at once. What the paths can do is cancel or reinforce each other before the one string is drawn.

2.4 HH versus the coin

Example 2.4 (Q: two Hadamards against two coin flips)**.** Since H is self-adjoint and unitary,



(2.6)

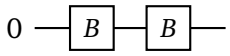
as matrices. Read it from $|0\rangle$ through the path sum, with $T=2$: a path is $(0,x_1,y)$, with amplitude $H_{yx_1}H_{x_10}$. The four paths, for $(x_1,y)=(0,0),(1,0),(0,1),(1,1)$, carry

$$\tfrac{1}{2},\quad \tfrac{1}{2},\quad \tfrac{1}{2},\quad -\tfrac{1}{2}.$$

The two paths to $y=0$ add to 1, and the two paths to $y=1$ cancel. The twin replaces each H by

$$|H|^{\circ 2}=\tfrac{1}{2}\begin{pmatrix}1&1\\1&1\end{pmatrix}=B,$$

the fair coin, which outputs a uniform bit whatever its input:



Its two flips give $y = 1$ with probability $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$, and $B^2 = B$: the coin again. The cross terms, $2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$ for $y = 0$ and $2 \cdot \frac{1}{2} \cdot (-\frac{1}{2}) = -\frac{1}{2}$ for $y = 1$, are the whole difference between the coin's $\frac{1}{2}, \frac{1}{2}$ and the qubit's $1, 0$.

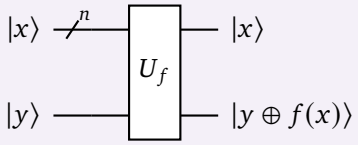
Read after one Hadamard, the qubit has the coin's statistics; read after two, it does not. And $|HH|^{\circ 2} = \mathbb{1}$, while $|H|^{\circ 2}\,|H|^{\circ 2} = B$: the squared modulus does not respect composition, as (iv) said.

2.5 Kickback and Deutsch

Definition 2.5 (Oracle, query). For $f\colon \{0,1\}^n \to \{0,1\}$, the *oracle* of f is the matrix U_f of Definition 1.8 on $n+1$ qubits, query register first,

$$U_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x)\rangle\,,$$

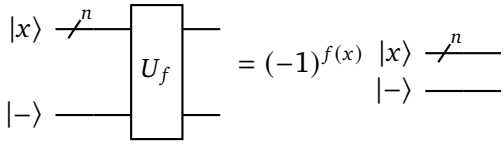
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A quantum procedure may apply U_f to any state of its $n+1$ qubits; a classical one presents a string (x, y) and receives $(x, y \oplus f(x))$, so it learns $f(x)$. Each use of U_f is one *query*.

The oracle U_f is a permutation matrix with real entries, so $U_f^\dagger U_f = U_f^{\mathsf{T}} U_f = \mathbb{1}$: it is a gate. By Theorem 1.9 it has a circuit with clean ancillas whenever f has a classical one; here only the box matters, and the question is how many queries a task needs.

Phase kickback. Feed the target the state $|-\rangle$. For every $x \in \{0,1\}^n$,



(2.7)

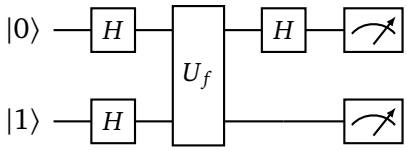
an equality of vectors: $U_f\,|x\rangle\,|-\rangle = (-1)^{f(x)}\,|x\rangle\,|-\rangle$. Indeed, since $\otimes$ is linear in each factor, (1.3) gives

$$|x\rangle\,|-\rangle = \frac{|x0\rangle - |x1\rangle}{\sqrt{2}},$$

and U_f sends $|x0\rangle$ to $|x\,f(x)\rangle$ and $|x1\rangle$ to $|x, 1 \oplus f(x)\rangle$. If $f(x) = 0$ it fixes both terms; if $f(x) = 1$ it exchanges them, which reverses the sign of their difference.

The value $f(x)$ is not written into the target, which leaves as $|-\rangle$, as it came. It comes back as a sign on the query register: it is *kicked back*. On one $|x\rangle$ the sign multiplies the whole state and changes no reading; on a superposition of several x the signs can differ from term to term, and interference can read them.

Example 2.6 (Q: Deutsch's problem)**.** Given the oracle of $f\colon \{0,1\} \to \{0,1\}$, decide whether f is constant: compute $b = f(0) \oplus f(1)$. The circuit



makes one query. Column by column: first $(H \otimes H)\,|0\rangle\,|1\rangle = |+\rangle\,|-\rangle$, by the interchange law. Then, by (2.7) on each term of $|+\rangle\,|-\rangle = \frac{1}{\sqrt{2}} \sum_x |x\rangle\,|-\rangle$,

$$\begin{aligned} U_f\,|+\rangle\,|-\rangle &= \frac{1}{\sqrt{2}} \sum_x (-1)^{f(x)}\,|x\rangle\,|-\rangle \\ &= (-1)^{f(0)}\,H\,|b\rangle\,|-\rangle\,, \end{aligned}$$

since $(-1)^{f(1)} = (-1)^{f(0)}(-1)^b$ and $H\,|b\rangle = (|0\rangle + (-1)^b\,|1\rangle)/\sqrt{2}$. Last, $H \otimes \mathbb{1}$, with $H^2 = \mathbb{1}$, leaves

$$(-1)^{f(0)}\,|b\rangle\,|-\rangle\,,$$

whose amplitudes are $\pm 1/\sqrt{2}$ on $b0$ and $b1$ and 0 elsewhere. Reading gives $b0$ or $b1$, each with probability $\frac{1}{2}$: the top bit is b with certainty, after one query.

The sign $(-1)^{f(0)}$ multiplies every amplitude, so it changes no probability of a reading; that it changes nothing at all, whatever follows, is proved in Lecture 3. The interference is in the last Hadamard. Before the reading the state is $\sum_c \alpha_c\,|c\rangle\,|-\rangle$, with $\alpha_c = \frac{1}{2}\big((-1)^{f(0)} + (-1)^{f(1)+c}\big)$, one term from each query value x. For $c = b$ the two terms are equal; for $c = 1 \oplus b$ they are opposite and cancel, as the paths to $y = 1$ did in Example 2.4.

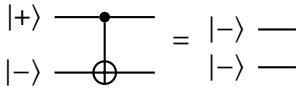
Proposition 2.7 (One classical query)**.** *Every classical procedure that queries $f\colon \{0,1\} \to \{0,1\}$ once, deterministic or with random bits, outputs $f(0) \oplus f(1)$ correctly with probability at most $\frac{1}{2}$ for at least one of the four f.*

Proof. Fix the random bits. The procedure then presents a fixed string (x, y), receives $y \oplus f(x)$, and answers with a function of $f(x)$ alone. The two f that agree at x differ in $f(0) \oplus f(1)$, so it is right for exactly one of each pair, that is, for two of the four f. Its success averaged over the four f is therefore $\frac{1}{2}$ for every value of the random bits, so its probability of success averaged over the four f is $\frac{1}{2}$, and for some f that probability is at most $\frac{1}{2}$. ∎

A fair guess succeeds with probability $\frac{1}{2}$ for every f. One classical query cannot improve that worst-case guarantee; two always succeed, while one quantum query suffices. Lecture 5 formalises this query comparison and gives a separation that grows exponentially with the number of input bits, relative to an oracle.

2.6 The control guard

Example 2.8 (QQ: the control changes)**.** For $f(x) = x$, the oracle $U_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus x\rangle$ is the CNOT of Definition 1.7. By the kickback on each term of $|+\rangle$,



(2.8)

an equality of vectors. The control enters as $|+\rangle$ and leaves as $|-\rangle$; the target is unchanged.

Lecture 1's rule that controls are read and left unchanged is a statement about basis states. On a superposition the control can change, and the kickback is exactly that change. Phase estimation (Lecture 7) is built on it: there the phase lands on the control register.

Summary.

- *Handed on:* qubits, gates and readings (Definition 2.1); a circuit has one unitary matrix, and its mirror image with adjoint boxes is the inverse (Theorem 2.2).
- *The main result:* an amplitude is a sum over paths; the probabilistic twin sums the squared moduli of path amplitudes; the cross terms are interference (Theorem 2.3 and Example 2.4).
- *Kickback:* on $|-\rangle$ the oracle puts $(-1)^{f(x)}$ on the query register; one query solves Deutsch's problem, and one classical query cannot improve a fair guess's worst-case guarantee (Example 2.6 and Proposition 2.7).
- *The guard:* on $|+\rangle\,|-\rangle$ the control of a CNOT changes (Example 2.8); Lecture 7 uses it.
- *Next:* Lecture 3, the state of part of a register.

References

QM-I, Lectures 1 and 4. QM-II, Lecture 7 (Hamiltonian evolution). Nielsen and Chuang, §§1.2–1.3.1, pp. 13–20, §§1.4.2–1.4.3, pp. 30–34, §2.1.6, pp. 69–70, and Ex. 4.20, pp. 179–180. Watrous (2025), §1.2, pp. 12–23, and §§5.1–5.2, pp. 131–137 (its U_f writes the target first, p. 137). De Wolf, pp. 17–18, and §13.3, pp. 112–113 (the path sum). Selinger, *A survey of graphical languages for monoidal categories* (2009), Thm 3.12, p. 18, and §7.2, pp. 50–51. Deutsch's problem is from D. Deutsch, Proc. R. Soc. Lond. A **400** (1985), 97–117; its one-query solution with certainty from R. Cleve, A. Ekert, C. Macchiavello and M. Mosca, Proc. R. Soc. Lond. A **454** (1998), 339–354.